Certified Floors on Decision Risk

Train-Free Lower Bounds on Bayes Risk from Outcome-Blind Near-Replicates

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Abstract

Model error confounds two quantities: information absent from an input and information present but unused. Only the second is reducible by better modelling. We introduce **Peras**, a train-free procedure for finite-design *lower* confidence bounds on Bayes risk under absolute error, pinball loss and within-window miss loss. Validity is conditional on an externally justified Lipschitz selection of Bayes actions, outcome-blind geometry and independent entity outcomes; it is not assumption-free. If nearby cases have different outcomes, a smooth Bayes action cannot explain both without error. An outcome-blind *group* witness uses a virtual common action to retain signal that pairs lose. On ten out-of-distribution MAE generators with analytic Bayes risk, grouped and paired bounds lie below truth in 200/200 draws; the group is strictly larger in 180/200, ties at zero in 20/200, and raises median usable tightness from 0.563 to 0.641. On a disjoint five-loss benchmark a frozen route improves W15 and W30 bounds but ties paired Bentkus on quantiles. We derive a universal $\tau(1-\tau)R$ cap for grouped pinball scores and a sharper fixed-group cap. Sparse high-dimensional inputs yield zero bounds, and external local methods beat the witness centre as a point estimator. A retrospective operating-room calculation illustrates the method but is *not* a clinical certificate: its action-smoothness and outcome-support assumptions remain unaudited.

1 Introduction

A predictive model on a benchmark reports one number: the error of that model. The number that governs whether further work is worthwhile is different and counterfactual — *how well could any decision rule have performed, given only the information available at decision time?* The gap between these two quantities is the reducible part of the error. The remainder is forced by the information set and cannot be engineered away.

The distinction is not academic. In operating-room scheduling, a minute of overrun attributable to an unread clinical note calls for a better model; a minute attributable to genuine outcome variability given the whole preoperative record calls for a different intervention entirely — wider blocks, different staffing, or no action at all. Practitioners routinely spend years pursuing headroom that does not exist, because nothing in standard evaluation tells them where the floor is.

Formally, for a declared pre-decision representation Z, outcome Y, and loss ℓ, the object of interest is the Bayes risk

$$R^\star_\ell(Z) \;=\; \mathbb{E}\Big[\inf_a \mathbb{E}\{\ell(Y,a) \mid Z\}\Big]. \tag{1}$$

This is a property of the joint law under a specified decision problem, not of any fitted model. Estimating it is hard: the infimum is over all measurable rules, and the conditional law of $Y \mid Z$ is unknown.

Our approach. We do not estimate $R^\star_\ell$; we *lower bound* it, with finite-sample confidence, without fitting any predictor. The mechanism is geometric. If two cases are close under a declared metric, a selected *Bayes action* is assumed to assign them nearly the same action; arbitrary decision rules need not. If their outcomes nonetheless differ by D, at least one Bayes action incurs loss. Formalising this costs one Lipschitz assumption and one triangle inequality, and it yields pointwise *witnesses* whose expectations lie below average conditional Bayes risk. Standard one-sided concentration converts witnesses into conditional certificates.

Relation to matching estimators. Matching, nearest-neighbour and constrained-action methods already address irreducible error [Devroye et al., 1996, Rice, 1984, Anderson & Bjarnadottir, 2024]. Our claim is narrower than a new matching principle: under explicit structural and sampling premises, a predeclared registry of pair and group witnesses gives a one-sided finite-design confidence statement across several decision losses. We have not implemented all published floor methods on identical data. The price is that this is a bound, not an accurate point estimator; external local methods locate the floor far more accurately than our witness centre (§8.4).

Contributions.

1. **A loss-indexed witness family** (§4) giving finite-sample lower bounds on Bayes risk for MAE, pinball at any τ, and within-window miss rate, conditional on a selected Lipschitz Bayes action. This combination is the object we study; a first-of-its-kind priority claim would require a broader literature audit.
2. **The group witness** (§5, Theorem 5), our main methodological contribution: an outcome-blind partition and a *virtual common action*, justified by a one-dimensional interval-intersection argument. It is strictly larger than the paired construction in 180/200 MAE benchmark draws and ties at zero in 20/200 (§8.1); this is not universal dominance.
3. **A universal concentration range** for grouped pinball scores, $\tau(1-\tau)R$, with a sharper finite-group refinement (Proposition 6); and a **cost-free monotone closure** across window widths (Proposition 7).
4. **A multiplicity-controlled certificate** (Theorem 8) over a predeclared registry of witnesses, clipping levels and concentration methods, with an explicit refusal contract.
5. **A non-identifiability counterexample** for disagreement alone (Proposition 9), explaining why a separate outcome-based witness is needed.
6. **Honest empirics**, including three negative results that delimit the method: adaptive family selection is dominated under multiplicity, the witness centre is a poor point estimator, and sparse high-dimensional geometry silences the bound while preserving validity.

2 Related Work

Bayes error estimation. A mature literature studies classification Bayes error, including k-NN bounds [Devroye et al., 1996] and direct estimators [Ishida et al., 2023]; the present witness family does not cover 0–1 loss. Regression difference estimators [Rice, 1984] target residual variance under smoothness. More directly related, Anderson & Bjarnadottir [2024] estimate possible prediction performance using globally constrained actions. We do not implement their exact protocol here, so our experimental paired baseline is not a comprehensive comparison with published Bayes-floor

methods. The distinction sought is a one-sided confidence statement for a fixed feature design and multiple decision losses, rather than an unqualified claim that matching is new.

Minimax and information-theoretic lower bounds. Bayes-risk lower bounds via van Trees and Le Cam [Van Trees, 1968, Chen et al., 2016] are minimax statements over a class, with sharp constants for weighted quadratic loss. They bound risk for a *family* of distributions; ours is read off a specific finite design. Information-theoretic limits on log-loss forecasting reduce to mutual information, whose lower estimation is known to require sample sizes exponential in its magnitude [McAllester & Stratos, 2020], which is precisely the route our construction avoids.

Conformal prediction and risk control. Conformal risk control [Angelopoulos et al., 2024] provides distribution-free guarantees on the risk of a *specified* procedure. Our estimand is the minimum risk over all measurable rules given a declared input. Comparing the numerical outputs as if they were the same object is a category error; the two are complementary, and we use conformal machinery nowhere in the bound.

Disagreement-based estimates. Generalisation–disagreement equality [Jiang et al., 2022] relates ensemble disagreement to test error under a calibration condition later shown to be essential [Kirsch & Gal, 2022]. Recent work bounds a target predictor's risk through a certifiable surrogate [Bazinet et al., 2026] or controls model agreement through anchoring [Eaton et al., 2026]; neither directly certifies the Bayes floor of a declared information set. An earlier version of Peras pursued disagreement-based floors. Proposition 9 gives the elementary obstruction: agreement does not distinguish correctness from a shared blind spot. These papers and Peras answer different questions, not a head-to-head benchmark on the same estimand.

3 Setup

3.1 Estimand

Let $(Z_i, Y_i)_{i=1}^n$ be sampling entities with covariates z_i and outcomes $Y_i \in \mathbb{R}$. Write $r_\ell(z) = \inf_a \mathbb{E}\{\ell(Y, a) \mid Z = z\}$ for the pointwise Bayes risk. We target the *finite-design* risk, averaged over the audited covariates:

$$R^\star_{\ell,n} \;=\; \frac{1}{n}\sum_{i=1}^{n} r_\ell(z_i). \tag{2}$$

All statements are conditional on $z_{1:n}$. This is deliberately narrower than (1): transfer to a future population requires a separate sampling argument, which we do not claim.

Assumption 1 (Conditional independence)**.** Distinct sampling entities have conditionally independent outcomes given $z_{1:n}$.

Assumption 2 (Outcome-blind design)**.** The metric d, the pair and group partitions, the loss set, the clipping levels, the concentration methods and the entire candidate registry are fixed without reference to $Y_{1:n}$.

Assumption 3 (Lipschitz Bayes action)**.** There exists a selection $a^\star(\cdot)$ of Bayes actions and a constant L with $|a^\star(z) - a^\star(z')| \le L\, d(z, z')$ for all audited z, z'.

Assumption 3 is a scientific claim about the problem, not a statistical one. It must be defended externally; estimating L from the same outcomes and then invoking the theorem voids the guarantee. We treat L as declared, and in §8.5 we report explicitly that our clinical L is illustrative rather than audited.

Proposition 1 (An external route to a quantile cap)**.** *Suppose each conditional cdf F_z is continuous with a unique τ-quantile $q_\tau(z)$; for audited z, z' and all t, $|F_z(t) - F_{z'}(t)| \le K d(z, z')$; and each conditional density is at least $m > 0$ between the two quantiles. Then $|q_\tau(z) - q_\tau(z')| \le (K/m) d(z, z')$.*

Indeed, if $q_\tau(z) < q_\tau(z')$, the density lower bound gives $F_z(q_\tau(z')) - F_z(q_\tau(z)) \ge m(q_\tau(z') - q_\tau(z))$, while the cdf discrepancy at $q_\tau(z')$ is at most $K d(z, z')$; reverse the labels for the other ordering. Thus a *separate* scientific or independent-data audit of K and m could supply $L = K/m$ for median or pinball decisions. We have no such audit for Mayo. A window-maximising centre need not be continuous when two modes exchange rank, so this proposition does not justify its L.

3.2 The three decision losses

- **Absolute error:** $\ell_{\mathrm{MAE}}(y, a) = |y - a|$; Bayes act: conditional median.
- **Pinball at level τ:** $\ell_\tau(y, a) = \tau(y - a)_+ + (1 - \tau)(a - y)_+$; Bayes act: conditional τ-quantile.
- **Window miss at half-width w:** $\ell_w(y, a) = \mathbf{1}\{|y - a| > w\}$; Bayes act: max-mass centre.

The window Bayes act is generally *not* the median. The witness inequality does not need an analytic formula for it, but Assumption 3 still demands that a selected max-mass centre vary smoothly. This can fail near a mode switch. The window risk inverts into an operational ceiling: a valid lower bound B_w on miss risk implies attainable within-window accuracy no greater than $100(1 - B_w)\%$ for the audited finite design.

Proposition 2 (Decision value is loss-indexed)**.** *For nested representations,*

$$I_\ell(Z_1; Y \mid Z_0) = R^\star_\ell(Z_0) - R^\star_\ell(Z_0, Z_1) \ge 0.$$

The ordering of I_ℓ across channels may differ across losses.

The inequality is the classical comparison-of-experiments statement [Blackwell, 1953] specialised to a loss; we record it because it disciplines interpretation. A representation's *measured* floor may rise when a channel is added, since the witness depends on geometry; the true $R^\star_\ell$ cannot. When a certificate moves the wrong way, the certificate moved.

4 Pair Witnesses

Theorem 3 (Paired lower witnesses)**.** *Let (i, j) be a disjoint pair, $D = |Y_i - Y_j|$, $d_{ij} = d(z_i, z_j)$. Under Assumptions 1–3,*

$$h^{\mathrm{MAE}}_{ij} = \tfrac{1}{2}\,[D - L d_{ij}]_+\,, \qquad \mathbb{E}[h^{\mathrm{MAE}}_{ij}] \le \tfrac{1}{2}\big(r_{\mathrm{MAE}}(z_i) + r_{\mathrm{MAE}}(z_j)\big), \tag{3}$$

$$h^{\tau}_{ij} = \tfrac{1}{2}\min(\tau, 1 - \tau)\,[D - L d_{ij}]_+\,, \qquad \mathbb{E}[h^{\tau}_{ij}] \le \tfrac{1}{2}\big(r_\tau(z_i) + r_\tau(z_j)\big), \tag{4}$$

$$h^{w}_{ij} = \tfrac{1}{2}\mathbf{1}\{D > 2w + L d_{ij}\}, \qquad \mathbb{E}[h^{w}_{ij}] \le \tfrac{1}{2}\big(r_w(z_i) + r_w(z_j)\big). \tag{5}$$

Proof sketch; full proof in Appendix A. Write $a_i = a^\star(z_i)$. The triangle inequality gives $D \le |Y_i - a_i| + |a_i - a_j| + |a_j - Y_j|$, so $|Y_i - a_i| + |Y_j - a_j| \ge D - |a_i - a_j| \ge D - Ld_{ij}$; the left side is nonnegative, so the clipped form holds pointwise, and taking conditional expectations gives (3). For (4), $\ell_\tau(y, a) \ge \min(\tau, 1-\tau)|y - a|$ pointwise. For (5), if both outcomes were captured then $D \le 2w + Ld_{ij}$; the contrapositive forces at least one miss. □

Remark 4. Clipping before the deterministic penalty preserves the inequality and bounds the score, which is what one-sided Hoeffding and empirical Bernstein require. Clipping levels are part of the registry and consume confidence budget.

5 The Group Witness

Pairs are a luxury of low dimension. As dimension grows, nearest-neighbour distances inflate, Ld_{ij} absorbs D, and every paired witness clips to zero. The floor is unchanged; the instrument has gone deaf. Our remedy replaces pairwise comparison with a *virtual common action* over a group.

Let $\mathcal{G}$ be a partition of the audited indices formed from $z_{1:n}$ alone, with each group G assigned a feature-measurable centre c_G (coordinate mean or coordinate median; both are outcome-blind). Define per-case radii $\rho_i = L\, d(z_i, c_G)$ and the *relaxed group score*

$$S_G \;=\; \min_{a \in \mathbb{R}} \; \frac{1}{|G|} \sum_{i \in G} \tilde{\ell}(Y_i, a; \rho_i), \tag{6}$$

where the relaxation is $\tilde{\ell}_{\mathrm{MAE}}(y, a; \rho) = [|y - a| - \rho]_+$, $\tilde{\ell}_\tau(y, a; \rho) = \tau\,[y - a - \rho]_+ + (1 - \tau)\,[a - y - \rho]_+$, and $\tilde{\ell}_w(y, a; \rho) = \mathbf{1}\{|y - a| > w + \rho\}$. The asymmetric pinball expression matches the implemented witness; replacing it with $\min(\tau, 1-\tau)\,[|y - a| - \rho]_+$ is valid but strictly weaker in general. For the window loss the minimisation in (6) is a maximum-overlap computation over expanded intervals and is solved exactly in $O(|G| \log |G|)$.

Theorem 5 (Group witness)**.** *Under Assumptions 1–3, for every group G,*

$$\mathbb{E}[S_G] \;\le\; \frac{1}{|G|} \sum_{i \in G} r_\ell(z_i).$$

Proof sketch; full proof in Appendix B. Consider the intervals $I_i = [a^\star(z_i) - \rho_i,\; a^\star(z_i) + \rho_i]$ for $i \in G$. For any $i, j \in G$, Assumption 3 and the metric triangle inequality give $|a^\star(z_i) - a^\star(z_j)| \le L\, d(z_i, z_j) \le L\, d(z_i, c_G) + L\, d(c_G, z_j) = \rho_i + \rho_j$, so $I_i \cap I_j \neq \emptyset$. Pairwise intersecting intervals on the line have a common point (Helly's theorem in dimension one); let $a^\dagger$ be one. Since $a^\dagger$ depends only on $z_{1:n}$, it is a fixed feasible action conditional on the design. For each i, $|a^\dagger - a^\star(z_i)| \le \rho_i$, hence $\tilde{\ell}(Y_i, a^\dagger; \rho_i) \le \ell(Y_i, a^\star(z_i))$ pointwise for each of the three relaxations. Therefore $S_G \le |G|^{-1} \sum_{i \in G} \tilde{\ell}(Y_i, a^\dagger; \rho_i) \le |G|^{-1} \sum_{i \in G} \ell(Y_i, a^\star(z_i))$, and taking conditional expectations gives the claim. □

Two properties matter. First, the bound holds for *any* feature-measurable centre; changing the centre changes witness strength, never the target or the confidence allocation. Second, disjoint groups have conditionally independent scores under Assumption 1, so group scores concentrate exactly as pair scores do.

5.1 Universal and finite-group ranges for pinball scores

Proposition 6 (Universal and finite-group pinball ranges)**.** *If outcomes lie in a declared interval of length R, then $0 \le S_G \le \tau(1-\tau)R$ for every group. For a fixed group size k, the sharper bound is*

$$S_G \;\le\; R \max_{m=0,\dots,k} \min\left\{\frac{\tau(k-m)}{k}, \frac{(1-\tau)m}{k}\right\}. \tag{7}$$

The finite-group constant is attainable when group features coincide ($L = 0$) and outcomes occupy the two support endpoints in the maximising proportions. The universal constant is sharp over group sizes, but need not be attained for a fixed k.

The universal cap improves the looser $\min(\tau, 1-\tau)R$ bound, by a factor of 2 at $\tau = 1/2$. The finite refinement is opt-in in the implementation so the historical default is unchanged. Both affect concentration, not the witness inequality; proof and examples are in the supplement.

5.2 Free monotone closure across windows

Proposition 7 (Window closure)**.** *$R^\star_w$ is nonincreasing in w. Hence for a predeclared grid, replacing B_w by $\max_{v \ge w} B_v$ preserves validity and can only tighten the bound at smaller widths, at no additional confidence cost.*

The closure is deterministic post-processing of already-valid bounds and spends no α. It cannot repair an unaudited L or a grid chosen after reading outcomes.

6 Certification

Theorem 8 (Simultaneous certificate)**.** *Let $\mathcal{H}$ be a registry — fixed under Assumption 2 — of candidate witnesses (pair or group), clipping levels, group sizes and one-sided concentration methods, with budgets $\alpha_h > 0$, $\sum_h \alpha_h \le \alpha$. If $\underline{\mu}_h$ is a valid conditional $(1-\alpha_h)$ lower confidence bound on $\mathbb{E}[h]$, then*

$$B_\ell = \max\left\{0,\ \max_{h \in \mathcal{H}} \underline{\mu}_h\right\} \quad \textit{satisfies} \quad \mathbb{P}\{B_\ell \le R^\star_{\ell,n} \mid z_{1:n}\} \;\ge\; 1-\alpha.$$

Proof. Each event $\{\underline{\mu}_h > \mathbb{E}[h]\}$ has probability at most α_h. Off their union, every $\underline{\mu}_h \le \mathbb{E}[h] \le R^\star_{\ell,n}$ by Theorem 3 or 5; the maximum of quantities below $R^\star_{\ell,n}$ is below $R^\star_{\ell,n}$, and $R^\star_{\ell,n} \ge 0$. A union bound gives the level. □

We implement Hoeffding, Maurer–Pontil empirical Bernstein [Maurer & Pontil, 2009], and Bentkus binomial-tail inversion [Bentkus, 2004] as interchangeable radii. Multiple losses, representations, slices or periods require an *outer* allocation of α; reusing an outcome to select any registry element voids the level. The procedure is summarised in Algorithm 1.

7 A Sensitivity Bridge to Disagreement Bounds

Ensemble disagreement is an appealing route to the same floor, but it cannot identify irreducible risk by itself.

Proposition 9 (Disagreement alone does not identify Bayes MAE)**.** *No function only of pairwise predictor disagreement can provide a universally positive lower bound on Bayes MAE. In particular, two identical constant-zero predictors have zero disagreement both when $Y = 0$ almost surely, with Bayes MAE 0, and when Y is equally likely to be $-M$ or M for any $M > 0$, with Bayes MAE M.*

Algorithm 1 Peras certificate for loss ℓ

Require: covariates $z_{1:n}$, outcomes $Y_{1:n}$, metric d, Lipschitz cap L, level α

Declare (outcome-blind): pair partition $\mathcal{P}$; group registry $\{\mathcal{G}_k\}$ with centres; clip grid $\mathcal{C}$; methods $\mathcal{M}$

$\mathcal{H} \leftarrow \mathcal{P} \times \mathcal{C} \times \mathcal{M} \cup \{\mathcal{G}_k\} \times \mathcal{C} \times \mathcal{M}$; assign $\alpha_h = \alpha/|\mathcal{H}|$

for $h \in \mathcal{H}$ **do**

    compute conditionally independent scores, which need not be identically distributed

    $\underline{\mu}_h \leftarrow$ one-sided $(1 - \alpha_h)$ lower confidence bound via method $m(h)$

end for

$B_\ell \leftarrow \max\{0, \max_h \underline{\mu}_h\}$; apply window closure (Prop. 7) if $\ell = \ell_w$

return B_ℓ with registry, budgets, L, metric, and provenance hashes; **refuse** if any field is absent

The two worlds have the same predictors, inputs and disagreement, but different floors. An additive shared-error sensitivity calculation could only become a Bayes-risk bound with an *independently valid* bridge from its error quantity to Bayes risk; subadditivity of MAD alone supplies no such bridge. The previous draft asserted that final implication without a premise, so we remove it rather than relabel it as proved. Peras therefore uses outcomes and a declared action structure in its primary construction.

8 Experiments

All generators are frozen before evaluation with analytic Bayes risks, verified by independent Monte Carlo and by exhaustive enumeration of two-point conditional laws. Development and evaluation use disjoint seeds. Peras fits no predictor; all point estimators are external comparators.

8.1 Main result: the group witness wins on this MAE benchmark

Ten out-of-distribution MAE generators — beta-Gaussian covariates, separated support with Laplace noise, oscillatory centre, V-shaped centre, radial 2D centre, 3D max centre, sparse 20D, heteroskedastic Gaussian, heteroskedastic Laplace, shifted exponential skew — supply analytic conditional medians, MAE floors and Lipschitz caps, evaluated at the *sampled* covariates. Seed `20260928`, 20 draws per family, $n = 1600$ entities per draw. Unbounded outcomes are handled by clipping on the frozen grid $(0.05, 0.1, 0.2, 0.4, 0.8, 1.6)$ with Bonferroni across the grid, methods, and group sizes.

Method	Valid	Median usable tightness	Paired wins
Clipped paired, 95%	200/200	0.5634	0
Clipped grouped, 95%	**200/200**	**0.6405**	**180**

The group witness wins *every* draw in nine of ten families and never loses; the twenty remaining draws are ties at zero on sparse 20D. The 200/200 coverage has Wilson lower limit 0.981. An independent earlier seed (`20260925`, 10 draws per family) reproduces the pattern exactly: 90 grouped wins, 0 paired wins, 10 ties. Usable tightness is $B/R^\star$ when valid and 0 when invalid, so a larger invalid bound scores zero and cannot be traded for tightness.

This is a within-benchmark result against the implemented paired baseline. It is not a claim of superiority over all published bounds, nor evidence that L is knowable on real data. Supplementary

Table S1 gives the analytic gold risk and both median bounds for every design, including the sparse failure.

8.2 Five losses, disjoint benchmark

A second, non-overlapping suite of ten designs — five numeric, five with deterministic pre-outcome templated note text stressing extraction, negation, scale, community structure and rare cues — was evaluated on untouched seed `20261006`: 20 draws per design, $n = 512$, five losses (q10, q90, W15, W30, W60) sharing each draw. That is 200 independent draws and 1,000 loss results, each loss receiving $\alpha = 0.01$ for joint 95% coverage.

Route	Valid / 1,000	Median usable tightness
Adaptive pair/group envelope	1,000	0.1581
Frozen loss-only route	**1,000**	**0.1843**
Paired Hoeffding	1,000	0.0000
Paired empirical Bernstein	1,000	0.0717
Paired Bentkus	1,000	0.1818

The frozen route is paired Bentkus for pinball and *grouped* Bentkus for windows. It improves W15 median tightness to 0.2678 from 0.2175, winning 177/200 draws, and W30 to 0.0747 from 0.0580, winning 127/200; on q10/q90 it ties the paired baseline at 0.1862 by construction. All 200 five-loss joint draws are valid (pooled Wilson lower limit 0.981); no single design's 20 draws support more than 0.839. A later outcome-blind geometry route yields additional synthetic wins (Supplementary Table S2), but its window threshold was adjusted after seeing retrospective Mayo results, so those Mayo outputs are development diagnostics only.

Negative result: adaptivity loses under multiplicity. The adaptive envelope, which selects across witness families, beats paired Bentkus in only 237 of 800 positive-floor cells and loses in 543; the frozen route beats the adaptive envelope in 304 cells, loses in 76 and ties in 420. Adaptive selection purchases coverage insurance that the frozen route does not need and pays for it in α. We report the frozen route as the recommended configuration.

8.3 Where the witness goes deaf

On the sparse four-dimensional design, median tightness is 0.0234 (adaptive) and 0.0311 (routed). On sparse 20D in the MAE suite, both methods return zero. The floor in those worlds is not smaller; the geometry is worse. When near-replicates are distant under the declared metric, $L\,d$ consumes the observed spread and the witness clips to nothing. In fixed dimension, nearest-neighbour distances can shrink with sample size; the problem is the sample size required under a poor high-dimensional metric. A valid zero is honest and uninformative. Characterising this transition in (sample size, dimension, metric, L) is an important open problem.

An outcome-blind centre pilot (seed `20261007`) confirms that the limitation is not the centre rule: a coordinate-median centre improved the numeric bound in 180/200 results and worsened none, but median tightness moved only from 0.18814 to 0.18866.

8.4 Negative control: the witness centre is not an estimator

Cross-fitted local plug-ins, trained *outside* Peras, locate the floor far more accurately than our witness centre (median absolute error to known floor):

Loss	Witness centre	kNN	Histogram	Kernel
q10 pinball	1.9729	0.1905	0.2726	**0.1389**
q90 pinball	1.9643	0.1963	0.2419	**0.1723**
W15 miss	0.2857	**0.0156**	0.0192	0.0176
W30 miss	0.1242	0.0192	0.0191	**0.0173**

In the MAE suite the same holds: leave-one-out kNN achieves median absolute error 0.0050 against the known floor and wins 142/200 draws over a local jackknife at 0.0079. This is a demarcation, not a defect. An interval-side certificate and a point estimate are different objects, and the certificate should never be advertised as the latter.

8.5 Retrospective clinical audit

We applied the *legacy radial/adaptive diagnostic*, not the later geometry route, to 10,528 earliest-per-patient operating-room cases across four chronological panels and two representations, using frozen out-of-fold predictions as achieved-risk comparators; no model was trained inside Peras. With an illustrative $L = 300$ and a $[0, 900]$-minute outcome cap:

Representation	B_{W30}	Attainable ceiling	Achieved
Tabular	0.2019	$\leq 79.81\%$	62.60%
Tabular + note values	0.1882	$\leq 81.18\%$	—

If the declared smoothness, support and sampling assumptions held, the first row would imply a within-30-minute ceiling of 79.81% for the tabular information set, compared with 62.60% achieved by the frozen MAE-selected OOF comparator. It does not establish that one fifth of future cases is intrinsically unschedulable, and the comparator is not shown to be deployed. The window grid would also imply strict fixed-design percentile floors ($p_{75} > 15$ and $p_{90} > 30$ minutes) under those same conditions. Grouped pinball diagnostics are 2.3704 (q10) and 4.5177 (q90) loss-minutes.

By Proposition 2 the lower numerical bound for the richer representation is *not* evidence that notes reduce information. All clinical numbers are **exploratory**: L and the outcome cap are illustrative, and patient independence, feature timing and retrospective reuse are not independently audited. No patient-level text or outcome appears in this paper.

9 Limitations

The Lipschitz cap is assumed, not estimated. It is the load-bearing assumption and the one most likely to fail silently on real data. We provide a refusal contract rather than an estimator, and we regard estimating L from the same outcomes as inadmissible.

Baselines are concentration methods on a shared witness, plus external point estimators — not a head-to-head against every published Bayes-error method. For 0–1 loss our family does not apply, and the mature classification literature [Devroye et al., 1996, Ishida et al., 2023] is the correct tool.

Tightness is not established. We prove validity, not optimality, and exhibit regimes with near-zero tightness. No minimax statement is claimed.

Coverage is demonstrated, not proved uniformly. 1,000/1,000 non-violations under known truth do not establish uniform coverage over unseen distributions, and the templated text is an integration test rather than evidence of NLP generality.

Clinical claims are retrospective. A frozen prospective study of 3,000 new eligible cases with patient-clustered inference is required before any confirmatory statement.

10 Conclusion

We give a train-free, finite-sample, multiplicity-controlled lower certificate on Bayes risk for absolute-error, quantile and within-window decisions, conditional on an externally justified Lipschitz Bayes action, outcome-blind design, independent entities and appropriate boundedness or clipping. The group witness — a virtual common action over an outcome-blind partition, justified by a one-dimensional interval argument — exceeds the paired baseline in 180 of 200 known-risk MAE draws and ties at zero in the remaining 20. We also delimit the method sharply: adaptive family selection loses under multiplicity on a separate benchmark, the witness centre is not a competitive estimator, and sparse geometry silences the bound while preserving its conditional validity.

The open problem we consider most important is the deaf regime. A witness that survives sparse high-dimensional geometry — through a predeclared neighbourhood or graph design with a finite-sample correction for overlapping pairs — would convert a valid-but-silent bound into an informative one, and would do so exactly where modern representations live.

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A Proof of Theorem 3

Fix a disjoint pair (i, j) and condition throughout on $z_{1:n}$. Write $a_i = a^\star(z_i)$, $a_j = a^\star(z_j)$, $D = |Y_i - Y_j|$, $d_{ij} = d(z_i, z_j)$.

Absolute error. By the triangle inequality, $D = |Y_i - Y_j| \le |Y_i - a_i| + |a_i - a_j| + |a_j - Y_j|$, hence $|Y_i - a_i| + |Y_j - a_j| \ \ge \ D - |a_i - a_j| \ \ge \ D - Ld_{ij}$ using Assumption 3. The left-hand side is nonnegative, so $|Y_i - a_i| + |Y_j - a_j| \ge [D - Ld_{ij}]_+$ pointwise. Dividing by 2 and taking conditional expectations, and using $\mathbb{E}[|Y_i - a_i| \mid z_i] = r_{\mathrm{MAE}}(z_i)$ by definition of the Bayes action, gives (3).

Pinball. For any y, a and $\tau \in (0, 1)$, $\ell_\tau(y, a) = \tau(y-a)_+ + (1-\tau)(a-y)_+ \ \ge \ \min(\tau, 1-\tau)\,|y-a|$, since exactly one of the two terms is active and each coefficient is at least $\min(\tau, 1-\tau)$. Applying this at (Y_i, a_i) and (Y_j, a_j) and invoking the absolute-error step yields (4).

Window. Suppose both outcomes are captured: $|Y_i - a_i| \le w$ and $|Y_j - a_j| \le w$. Then $D \le |Y_i - a_i| + |a_i - a_j| + |a_j - Y_j| \le 2w + Ld_{ij}$. Contrapositively, $D > 2w + Ld_{ij}$ implies $\mathbf{1}\{|Y_i - a_i| > w\} + \mathbf{1}\{|Y_j - a_j| > w\} \ge 1$. Halving and taking expectations gives (5). □

B Proof of Theorem 5

Condition on $z_{1:n}$ and fix a group G with feature-measurable centre c_G and radii $\rho_i = L\,d(z_i, c_G)$.

Step 1: a common feasible action exists. For $i, j \in G$, Assumption 3 and the triangle inequality for d give

$$|a^\star(z_i) - a^\star(z_j)| \ \le \ L\,d(z_i, z_j) \ \le \ L\,d(z_i, c_G) + L\,d(c_G, z_j) \ = \ \rho_i + \rho_j,$$

so the closed intervals $I_i = [a^\star(z_i) - \rho_i,\ a^\star(z_i) + \rho_i]$ intersect pairwise. Helly's theorem in dimension one states that a finite family of pairwise intersecting intervals has a common point; let $a^\dagger \in \bigcap_{i \in G} I_i$. Because $a^\star(\cdot)$, c_G and d are functions of covariates alone, $a^\dagger$ is measurable with respect to $z_{1:n}$ and is therefore a fixed scalar under the conditioning.

Step 2: the relaxation is dominated pointwise. By construction $|a^\dagger - a^\star(z_i)| \le \rho_i$ for every $i \in G$. For absolute error, $|Y_i - a^\dagger| \le |Y_i - a^\star(z_i)| + \rho_i$, so $\Big[|Y_i - a^\dagger| - \rho_i\Big]_+ \le |Y_i - a^\star(z_i)|$. For pinball, if $Y_i - a^\dagger - \rho_i > 0$ then $Y_i - a^\star(z_i) \ge Y_i - a^\dagger - \rho_i > 0$; likewise, if $a^\dagger - Y_i - \rho_i > 0$ then $a^\star(z_i) - Y_i \ge a^\dagger - Y_i - \rho_i > 0$. Multiplying the two positive parts by τ and $1 - \tau$, respectively, shows $\tilde{\ell}_\tau(Y_i, a^\dagger; \rho_i) \le \ell_\tau(Y_i, a^\star(z_i))$. For the window loss, if $|Y_i - a^\dagger| > w + \rho_i$ then $|Y_i - a^\star(z_i)| \ge |Y_i - a^\dagger| - \rho_i > w$, so $\mathbf{1}\{|Y_i - a^\dagger| > w + \rho_i\} \le \mathbf{1}\{|Y_i - a^\star(z_i)| > w\}$.

Step 3: minimisation and expectation. Since $a^\dagger$ is feasible in (6),

$$S_G \ \le \ \frac{1}{|G|} \sum_{i \in G} \tilde{\ell}(Y_i, a^\dagger; \rho_i) \ \le \ \frac{1}{|G|} \sum_{i \in G} \ell(Y_i, a^\star(z_i)).$$

Taking conditional expectations and using $\mathbb{E}[\ell(Y_i, a^\star(z_i)) \mid z_i] = r_\ell(z_i)$ gives

$$\mathbb{E}[S_G] \le |G|^{-1} \sum_{i \in G} r_\ell(z_i).$$

Averaging over disjoint groups yields a lower witness for $R^\star_{\ell,n}$; disjointness and Assumption 1 make the S_G conditionally independent. □

C Proof of Proposition 6

Let outcomes lie in $[y_{\min}, y_{\min} + R]$ and set $a = y_{\min} + \tau R$. If $y \geq a$ then $\ell_\tau(y, a) = \tau(y - a) \leq \tau(y_{\min} + R - y_{\min} - \tau R) = \tau(1 - \tau)R$. If $y < a$ then $\ell_\tau(y, a) = (1 - \tau)(a - y) \leq (1 - \tau)(y_{\min} + \tau R - y_{\min}) = \tau(1 - \tau)R$. Hence every admissible observation has pinball loss at most $\tau(1 - \tau)R$ at this constant action, and since S_G minimises a relaxed score over all actions — and relaxation only decreases the score — $S_G \leq \tau(1 - \tau)R$.

For fixed $k = |G|$, relaxation gives $S_G \leq \min_a k^{-1} \sum_{i=1}^k \ell_\tau(y_i, a)$. The right side is convex in the vector $y_{1:k}$: the pinball objective is jointly convex in (y, a) and partial minimisation preserves convexity. Its maximum over the outcome-support cube is attained at a vertex. At a vertex with m lower and $k-m$ upper endpoints, choosing either endpoint action yields the smaller of $\tau(k-m)R/k$ and $(1-\tau)mR/k$; no interior action lowers this piecewise-linear objective further. Maximising over m proves (7). If all group features coincide and $L = 0$, the relaxation vanishes, so the maximising endpoint vector attains the finite cap. As k increases, proportions m/k can approach τ, making the finite constant approach $\tau(1 - \tau)R$. A two-atom population law with lower-atom mass τ attains this limiting Bayes pinball risk, but need not attain the empirical score cap for every finite k. □

D Proof of Proposition 7

For $w_1 < w_2$ and any action a, $\mathbf{1}\{|y - a| > w_2\} \leq \mathbf{1}\{|y - a| > w_1\}$ pointwise, so $r_{w_2}(z) \leq r_{w_1}(z)$ for every z and hence $R^\star_{w_2,n} \leq R^\star_{w_1,n}$. If the grid bounds are simultaneously valid — which Theorem 8 provides under a single outer allocation — then on the good event $B_v \leq R^\star_{v,n} \leq R^\star_{w,n}$ for every $v \geq w$, so $\max_{v \geq w} B_v \leq R^\star_{w,n}$. The closure is a deterministic function of already-certified quantities and therefore consumes no additional budget. □

Supplement to "Certified Floors on Decision Risk"

Proof Scope, Known-Risk Wins, Negative Controls, and Reproduction

Peras research working supplement

14 September 2026

Status. This is an editable research supplement to the revised reader draft, not an anonymized or template-compliant ICML submission. All Mayo quantities below are retrospective and exploratory. Synthetic values have analytic Bayes-risk targets. "Best" always means best among the named methods on the named frozen benchmark, never best among all published methods.

1 The exact scope of the certificate

Fix covariates $z_{1:n}$, disjoint outcome-blind groups, conditionally independent entity outcomes, a metric that extends to any virtual group centre, and a selected loss-specific Bayes action $a^*(z)$ satisfying $|a^*(z) - a^*(z')| \le Ld(z, z')$ on the audited design. The pair and group statistics are pointwise no larger than the loss of these Bayes actions. Their conditional expected scores therefore lie below $n^{-1}\sum_i r_\ell(z_i)$. One-sided concentration plus a union bound over the predeclared candidate registry yields a conditional finite-design lower confidence bound. It is not automatically a future-population or clinical guarantee.

What an external L audit would require. For median and pinball, suppose conditional cdfs obey $\sup_t |F_z(t) - F_{z'}(t)| \le Kd(z, z')$ and densities are at least $m > 0$ between their unique τ-quantiles. Then $|q_\tau(z) - q_\tau(z')| \le (K/m)d(z, z')$ by cdf inversion. An independent scientific or held-out-reference audit of K and m could justify $L = K/m$; choosing L after reading the evaluation outcomes cannot. No such bounds exist yet for Mayo. The max-mass centre for a window loss may jump discontinuously at a mode switch, so the quantile argument does *not* transfer to W30. Until an external window-action cap is available, its Mayo bound is only a numerical sensitivity diagnostic.

Implemented pinball score. The implemented radial group relaxation is asymmetric:

$$\tilde{\ell}_\tau(y, a; \rho) = \tau[y - a - \rho]_+ + (1 - \tau)[a - y - \rho]_+.$$

If $|a^\dagger - a^*(z_i)| \le \rho_i$, each of the two positive parts is bounded by the corresponding positive part of the true residual. Thus the relaxed pinball score is at most true pinball loss pointwise. The previous main-text formula $\min(\tau, 1 - \tau)[|y - a| - \rho]_+$ described a valid but weaker statistic and did not match code; the revised manuscript corrects it.

Fixed-group range. For outcomes in a population-valid support interval of span R and group size k, the exact worst-case *unrelaxed* empirical pinball optimum is

$$C_{k,\tau} = R \max_{m=0,\dots,k} \min\left\{\frac{\tau(k-m)}{k}, \frac{(1-\tau)m}{k}\right\}.$$

The relaxed score is no larger. The empirical optimum is a convex function of the outcome vector, so its maximum over the support cube occurs at an endpoint vector; m counts lower endpoints. Coincident group features and $L = 0$ attain the constant. For $\tau = 0.1$, $C_{2,\tau}/R = 0.05$, $C_{4,\tau}/R = 0.075$, and $C_{10,\tau}/R = 0.09$. The simpler $\tau(1-\tau)R$ is a valid universal cap and sharp as group size varies, but it is not necessarily attained by a *fixed* finite group. The earlier judge's sharpness approval applied to the two-atom population law, not to every empirical group size.

Shared-error correction. Subadditivity of an error dispersion such as MAD does not, by itself, imply a lower bound on Bayes risk. Identical zero predictors have the same zero disagreement when $Y = 0$ almost surely (Bayes MAE 0) and when Y is equally likely to be $-M$ or M (Bayes MAE M). The revised paper replaces the unsupported shared-bias-to-Bayes-risk proposition with this non-identifiability counterexample.

2 Known-risk MAE benchmark: every design, not just the headline

The frozen seed is 20260928; ten analytic-noise designs, 20 independent draws each, 1,600 entities per draw. Each entry in Table S1 is a median over that design's 20 draws. The gold column is the median *analytic conditional Bayes MAE at the sampled covariates*; the method columns are median one-sided nominal-95% bounds. Group and pair use their own Bonferroni-controlled clip/method registries. The final column counts strict group wins out of 20. Bold identifies the larger bound within each row; the gold is a target, not a competitor. All 200 grouped and paired bounds were below analytic truth in their draws; the draw-level 95% Wilson lower limit for 200/200 is 0.9812. This empirical count does not prove uniform coverage.

Table S1: Analytic gold and two train-free MAE bounds (medians; larger valid bound is tighter).

Design	Gold R^*	Paired	Grouped	Group wins
Beta–Gaussian	**0.2394**	0.1413	**0.1605**	20/20
Separated Laplace	**0.2000**	0.1210	**0.1360**	20/20
Oscillatory centre	**0.2000**	0.1002	**0.1162**	20/20
V-shaped centre	**0.1995**	0.1205	**0.1365**	20/20
Radial 2D	**0.1500**	0.0807	**0.0855**	20/20
3D maximum	**0.2500**	0.0969	**0.1053**	20/20
Sparse 20D	**0.2394**	**0.0000**	**0.0000**	0/20
Heteroskedastic Gaussian	**0.3192**	0.1818	**0.2084**	20/20
Heteroskedastic Laplace	**0.2996**	0.1743	**0.1903**	20/20
Shifted exponential	**0.1386**	0.0767	**0.0917**	20/20

Pooled median usable tightness B/R^* is 0.5634 paired and 0.6405 grouped; 180 group wins, zero pair wins and 20 ties at zero. Medians in Table S1 are not themselves valid confidence bounds and should not be divided to reconstruct the pooled ratio. A second seed, 20260925 with ten draws per family, gave 90 group wins, zero pair wins and ten sparse ties. The companion point-estimation negative control is important: leave-one-out external kNN had median absolute floor error 0.0050 versus 0.0079 for the train-free local jackknife and 0.0502 for the raw grouped centre. Peras wins this *bound* comparison, not the point-estimator comparison.

3 Multi-loss results and the changed route

The disjoint five-loss registry comprises q10/q90 pinball and W15/W30/W60 miss risk on ten duration-like designs (five numeric, five deterministic pre-outcome note templates). For the earlier frozen loss-only route, seed 20261006 supplied 20 draws/design at 512 entities/draw. The route used paired Bentkus for quantiles and grouped Bentkus for windows, allocating $\alpha = 0.01$ to each of the five losses per draw. All five losses were below truth jointly in 200/200 draws. On quantiles the route *ties* paired Bentkus by construction, whereas it improves W15 and W30 in 177/200 and 127/200 draws respectively. W60 has zero Bayes miss floor throughout this registry and contributes uninformative ties.

Table S2: Later geometry route on a fresh synthetic seed; comparison is numerical, not a jointly adjusted comparator certificate.

Quantity	Value
Seed; registered cells	20261015; 10 designs × 10 draws × 5 losses = 500
Cells below analytic truth	500/500
Route vs paired-classic maximum	339 wins / 140 ties / 21 losses
Median tightness, 400 positive-floor cells	0.2195 route / 0.1791 paired maximum
Design-level medians	9 strict route wins / 1 sparse-4D tie

Table S2 is an archived aggregate in `docs/MULTILOSS_VALIDATION_20260913.md`, not an independently rerun per-cell release artifact. Its paired comparator is a numerical maximum of three individually valid methods at the same alpha, *not* itself a multiplicity-adjusted 95% certificate. The route's window threshold was changed after inspecting Mayo retrospective results; the fresh synthetic seed provides a new synthetic check but cannot turn the reused Mayo cohort into confirmation. Exact-LP pinball outputs remain exploratory pending independent numerical verification.

4 Retrospective Mayo is an application, not the win claim

All entries use the same 10,528 earliest-per-patient cases in four chronological panels, with illustrative $L = 300$ and a claimed population support span of 900 minutes that has not been externally established. The main manuscript's W30 table is the *legacy radial/adaptive* calculation, not the later geometry route. The frozen OOF comparator is MAE-selected retrospectively and is not established as a deployed scheduling model.

Table S3: Exploratory Mayo sensitivity (not a clinical certificate).

Route	Representation	q90 pinball lower	W30 miss lower
Legacy radial/adaptive	Tabular	4.5177	0.2019
Legacy radial/adaptive	Tabular + note values	–	0.1882
Later geometry route	Tabular	5.2664	0.2678
Later geometry route	Tabular + note values	4.9176	0.2626

If all assumptions held, the legacy tabular W30 value would cap finite-design within-30-minute accuracy at 79.81%, compared with 62.60% for the frozen OOF comparator; the later geometry value would imply 73.22%. Neither is confirmed. The richer representation's smaller numeric bound does not imply that notes reduce Bayes information. A train-free 256-patient note corruption diagnostic also *failed* monotonicity: full note dropout lowered the W30 bound from 0.2574 to 0.2261 but raised q90 from 1.6756 to 1.7813, because the hashed-text metric and grouping changed. A separate actual-duration injection produced zero oracle error and a public certification refusal. No patient text or row-level outcome is reproduced here.

5 Reproduction and release ledger

The source, synthetic DGPs, comparator and tests are in the canonical project. From its root:

```
.venv/bin/python -m pytest -q -p no:cacheprovider
.venv/bin/python -c 'from horos.benchmark.ood_mae_comparison import \
run_ood_mae_comparison; print(run_ood_mae_comparison(\
seed=20260928,repetitions=20,n_entities=1600)["by_design"])'
.venv/bin/python scripts/run_mayo_text_augmentation.py --n-entities 256
```

The MAE benchmark reports `median_analytic_bayes_mae`, each method's `median_bound`, validity counts and paired wins. Mayo files are private and excluded from Git. The table values are aggregate-only. For the five-loss comparison and later geometry route, see `docs/MULTILOSS_VALIDATION_20260913.md`; a per-cell frozen artifact and independent external reproduction remain needed before a paper-level "best method" claim.

Claim tier	Current support
Conditional theorem	Finite-design witness and registry concentration, if smoothness, independence, support/clipping and outcome-blindness hold.
Known-risk simulation	MAE group wins 180/200 against the named paired baseline; newer geometry route wins 339/500 cells against a named numerical paired maximum.
Exploratory Mayo	All multi-loss clinical figures; L, window-action smoothness, support and temporal reuse unaudited.
Unsupported	Universal tightest bound, best point estimator, prospective clinical ceiling, exact-LP numerical proof, autonomous Poros.